Marketization of Home Production and Female Labor Supply*

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Abstract

This paper investigates the macroeconomic effects of a 25 percent reduction of taxes on market substitutes for home production, and how these effects build up over the life cycle. Using a multi-sector, overlapping generations collective household model calibrated to Germany, I incorporate dynamic human capital accumulation and intra-household bargaining. In the calibrated model, the reform reallocates activity toward the market sectors, especially services. Market services output expands by 8 percent and goods output by 1.4 percent. Driven by women's comparative advantage in services, the marketization of domestic work raises aggregate female market hours by 5.8 percent, against 4.1 percent for men, and increases the ratio of women's to men's market hours by about 1.6 percent. Because the calibration is still preliminary, these magnitudes are best read as provisional. These responses operate almost entirely through hours rather than wages, and they accumulate over the life cycle, partially offsetting the gradual specialization within the household.

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1 Introduction

Women work fewer hours in the market and more hours at home than men, a gap that has narrowed over time but remains substantial across advanced economies. In Europe, women devote on average around 26 hours per week to market work and 31 to home production, compared with 37 and 16 for men [1]. Raising female labor supply is a long-standing policy priority, motivated by gender equality, shortages of skilled workers, and the fiscal pressure of demographic change. Existing policy interventions have primarily taken the form of childcare subsidies, parental leave, and support for part-time work.

A distinct direction is to encourage the marketization of home production by lowering the cost of market substitutes for home-produced services (Ngai and Petrongolo (2017); Duval-Hernández, Fang, and Ngai (2023)). Many of these services, such as cleaning, cooking, and childcare, can be supplied through the market, and they are performed predominantly by women, in part because of a comparative advantage in producing services (Ngai and Petrongolo (2017)). A gender-neutral policy of this kind can therefore affect women and men very differently. Yet several of its implications remain underexplored. In particular, little is known about how its effects build up over the life cycle, as changes in labor supply feed into human capital accumulation and future wages, or about how it reshapes the balance of bargaining power within the household. In this paper I ask two questions. First, how do policies that promote the marketization of home production affect gender gaps in time use and wages? Second, how do these effects vary over the life cycle?

I begin by documenting how time use differs by gender over the life cycle in Germany. Using the German Socio-Economic Panel (SOEP), I estimate life-cycle profiles of market work, home production, and leisure for couples aged 25 to 54. These profiles show that gender differences in time allocation are not constant but evolve systematically over the working life, especially during child-rearing years, which both motivates a life-cycle approach and provides the empirical targets used to discipline the model.

To study the effects of the policy, I build an overlapping generations, multi-sector

[1] Eurostat (2025) and OECD (2021b).

general equilibrium model with home production and a collective household. The economy has two market sectors, one producing goods and one producing services, which are close substitutes for home-produced services. Each household consists of a woman and a man who decide jointly on consumption and on how to allocate their time across market work, home production, and leisure. Women have a comparative advantage in producing services, both at home and in the market. Two features make individual decisions dynamic. First, human capital accumulates with market experience, so current labor supply shapes future wages. Second, the household operates under limited commitment, and bargaining weights evolve endogenously as each spouse's outside option changes. The model adds three empirically motivated features: home production needs that vary with age, a human capital technology that is higher in the early periods of the life-cycle, and household specialization that emerges gradually. A government taxes labor income and consumption and rebates the revenue as a lump-sum transfer. The policy I consider is a reduction in the consumption tax on substitutable market services.

I calibrate the life-cycle model to the German economy. A subset of parameters is set externally from the literature. The remaining parameters are chosen so that the model reproduces the life-cycle profiles of wages estimated from the SOEP, together with aggregate moments such as the relative size of the different sectors and gender ratios in time use. The estimated profiles therefore serve a double role, documenting the patterns the model must explain and disciplining its key parameters.

I use the calibrated model to evaluate a 25 percent reduction in the consumption tax on substitutable market services. The policy works through several channels: a lower relative price of market services leads households to substitute market for home-produced services; the resulting reallocation of time changes market hours and, through experience, human capital and future wages. Because women dedicate a higher share of their time to both home and market services, these channels are expected to have a stronger effect on them. I find that reducing the tax on substitutable market services expands market services output by about 8 percent and raises aggregate women's

market hours by 5.8 percent, compared with 4.1 percent for men. This reallocation increases the ratio of women's to men's market hours by about 1.6 percent, while the aggregate impact on the gender wage ratio remains quantitatively negligible.

The remainder of the paper is organized as follows. Section 2 discusses the relevant literature and the contribution. Section 3 presents the empirical analysis of time use over the life cycle. Sections 4 develops the overlapping generations model. Section 5 describes the calibration, and Section 6 reports the quantitative results and counterfactuals. Finally, Section 7 concludes.

2 Related Literature

This paper speaks to three strands of the literature. First, it contributes to the work that uses multi-sector general equilibrium models to link market hours with the rise of services and the marketization of home production. One branch explains cross-country differences in working hours through tax and subsidy policy and the substitutability between market and home production. Rogerson (2007) shows that when revenues from higher taxes finance market substitutes of home produced services, such as childcare, part of the negative effect of taxation on market hours is offset, which helps reconcile high taxes with high market hours in Scandinavia. Olovsson (2009) finds that the gap in market hours between Sweden and the United States largely disappears once home production is accounted for, and Ngai and Pissarides (2011) and Ragan (2013b) attribute much of the variation in market work across OECD countries to differences in taxes, subsidies, and home-market substitutability. Rogerson (2008) links the US-Europe hours gap to the slower development of the European service sector. However, Duernecker and Herrendorf (2018) show that, following a tax increase, households substitute leisure for market work rather than reallocating time to home production. They argue a reason for this is the strong growth seen in the productivity of household production. Focusing on a single country, Ngai and Pissarides (2008) connect structural transformation to the U-shaped path of US hours: low cross-sector substitutability initially pulls labor into slower-growth services, while high substitutability between home and

market production eventually shifts labor from home to market work. Ragan (2013a) and Olovsson (2015) use related frameworks to study optimal fiscal policy and stress that home production and the substitutability between market and home services are central to policy design.

A subset of this literature focuses on gender differences, as does this paper. Freeman and Schettkat (2005) attribute cross-country differences in market work mainly to women's labor supply and to the higher marketization of home production in the United States. Olivetti and Petrongolo (2014) link cross-country variation in gender gaps in hours and wages to industry structure, in particular the size of the service sector. Ngai and Petrongolo (2017) show that structural transformation and marketization, both gender-neutral forces, raise women's relative wages and hours by expanding the service sector, where women hold a comparative advantage. Closest to the present paper are Rendall (2018) and Duval-Hernández, Fang, and Ngai (2023), both of which study the interaction between female labor supply and the expansion of the service sector. Rendall (2018) estimates how female labor supply feeds back into service-sector growth and finds that this channel accounts for more than one-third of that growth, driven mainly by married women, with higher and more progressive taxes reducing both female labor supply and the relative size of the service sector. Duval-Hernández, Fang, and Ngai (2023) document a negative cross-country correlation between gender gaps in hours and wages and show, in a multi-sector general equilibrium model with home production, that taxes and subsidies on market substitutes for household services explain a substantial share of it, reflecting women's comparative advantage in services. Building on these mechanisms — the substitution between home and market services and women's comparative advantage in producing these services — I analyze how a policy that lowers taxes on market substitutes for home production affects gender gaps in market hours and wages. I extend prior work in two directions. First, I add a life-cycle dimension to study how the policy's effects vary over the working life. Because household specialization evolves gradually after marriage (Mazzocco, Ruiz, and Yamaguchi, 2013) and human capital accumulation links current labor supply to future wages, the

timing of the policy matters and even small induced changes can have persistent effects. Second, I let the policy reshape the balance of power between spouses, a margin the literature above abstracts from, since shifts in relative labor market opportunities affect bargaining weights and, through them, labor supply and the allocation of household resources.

A second strand builds life-cycle models to study how taxes, transfers, and family policies shape labor market outcomes and gender gaps, emphasizing that labor supply decisions depend on dynamic considerations such as human capital accumulation, savings, and intra-household bargaining. On human capital, Blundell, Costa Dias, Meghir, and Shaw (2016) use UK reforms and a life-cycle model in which women choose labor supply, savings, and human capital to show that, because returns to part-time experience are low for less-educated women, tax credits raise employment in the short run but barely affect long-term wages. Adda, Dustmann, and Stevens (2017) estimate a life-cycle model on German data and find that most of the career cost of children stems from time out of work, with skill depreciation highest around mid-career. A related set of papers uses collective models to capture intra-household decision-making.[2] Mazzocco, Ruiz, and Yamaguchi (2013) show that household specialization evolves gradually after marriage, driven by childbirth, human capital accumulation, and limited commitment between spouses. Bronson, Haanwinckel, and Mazzocco (2024) examine how alternative tax systems affect household decisions and welfare, while Turon (2019) studies the trade-off between home and market childcare and shows that reducing labor supply to provide childcare at home lowers future earnings and, modestly, bargaining weights. Domeij and Klein (2012) and Bick (2016) use calibrated life-cycle models for Germany to evaluate childcare subsidies and find positive but limited effects on maternal labor supply, concentrated on the intensive margin. I contribute to this literature in two ways. First, I adopt a multi-sector perspective and explicitly model women's comparative advantage in service production, since women allocate substantially more time to services both in the market and at home, so policies that expand the service sector affect men

[2]See Chiappori and Mazzocco (2017), Section 3, for a discussion of intertemporal collective household models.

and women differently. Second, I incorporate general equilibrium effects, which allow me to capture the feedback from individual responses to aggregate outcomes. This is particularly relevant given Rendall (2018)'s finding that these feedback mechanisms can have substantial aggregate consequences.

Finally, the paper relates to the largely empirical literature on the effects of reducing taxes on market substitutes for home production. Flipo, Fougere, and Olier (2007) find that a French tax reduction on in-home services raised purchases of those services, mostly among higher-income households. Rickne (2021) studies the 2007 Swedish deduction for domestic services and finds employment gains concentrated among low-educated workers, with consumption still concentrated in high-income households. Examining the Belgian service voucher scheme, both Raz-Yurovich and Marx (2019) and Leduc and Tojerow (2024) find positive effects on female labor force participation and employment, across both the affected industries and the broader economy, for low- and high-skilled women alike. The fact that the effects extend beyond directly affected industries suggests the presence of important general equilibrium effects. A large related literature studies childcare subsidies, which can be read as a negative tax on market-provided childcare. For Germany, Bauernschuster and Schlotter (2015), Müller and Wrohlich (2020), and Zimmert (2023) find that expanding subsidized childcare raises maternal employment and hours, mainly through longer part-time jobs and among medium- or higher-educated mothers, whereas Gathmann and Sass (2018) show that a subsidy for families not using public care increased home childcare without raising overall parental labor supply. Looking at the long run, Lefebvre, Merrigan, and Verstraete (2009) use the 1997 Quebec reform to show that universal childcare produced lasting increases in maternal participation and hours, driven by less-educated mothers, and Morrissey (2017) surveys this evidence and concludes that lower out-of-pocket costs generally raise maternal employment, though estimated elasticities vary widely. I complement these studies by developing a structural model that allows me to identify and quantify the underlying mechanisms, incorporate general equilibrium effects, and perform counterfactual analysis.

3 Time Use over the Life Cycle

This section documents how the allocation of time across market work, home production, and leisure differs by gender for working-age individuals in Germany. These patterns motivate the life-cycle structure of the model and provide empirical targets used to discipline the parameters in Section 5.

3.1 Data

I use data from the German Socio-Economic Panel (SOEP), a large, nationally representative, longitudinal survey of households in Germany that has been conducted annually since 1984 by the German Institute for Economic Research (DIW) (Goebel, Grabka, Liebig, Kroh, Richter, Schröder, and Schupp, 2019). Each year, all adult members of participating households are asked about a wide range of social science topics, including household composition, occupation, employment, earnings, time use, health, and life satisfaction. For the analysis, I focus on variables related to time use, household composition, demographic characteristics of respondents, and labor market outcomes.

The time-use questions have been asked in a consistent form since the start of the panel. Respondents report the number of hours they spend on a typical weekday, and in a subset of waves on Saturdays and Sundays, across a fixed set of activities: paid work; errands; housework, such as washing, cooking, and cleaning; childcare; education and training; repairs and gardening; and hobbies and other leisure activities. I map these activities to the three uses of time in the model. Market work corresponds to hours in paid work, including apprenticeship, second jobs, and travel time to and from work. Home production aggregates the activities that have close market substitutes, namely housework, childcare, errands, and repairs and gardening. I consider leisure as the residual of a 168-hour weekly endowment to be consistent with the model.

To split market work into the two market sectors of the model, I follow Duval-Hernández, Fang, and Ngai (2023) and classify industries into goods and services accordingly [3]. For the wage analysis, I compute hourly wages from reported gross labor

[3] See Table A2 in Duval-Hernández, Fang, and Ngai (2023) for the detailed mapping.

earnings and hours worked, and I trim the top and bottom percentile of hourly wages within the estimation sample to limit the influence of measurement error in reported hours. To ensure comparability across time, all nominal values are deflated to constant 2015 euros using the Consumer Price Index (CPI) for Germany [4]. I restrict the sample to married and cohabiting individuals between 25 and 54 years old, therefore restricting attention to those who have generally completed their education but are not yet close to retirement. Additionally, I exclude individuals who report more than 112 hours per week (equivalent to 16 hours per day) of total home-production time and those with missing data on time-use variables, gender, educational attainment, or employment status. As the last restriction, I keep individuals observed at least twice after all other restrictions are imposed, so that the person fixed effects below are identified within the estimation sample. The final sample comprises around 27,000 individuals and 16,500 households, observed in all odd years from 1993 to 2021. This restriction for the years is due to the fact that time-use questions on weekend days are administered only every other year.

3.2 Life-Cycle Profiles

I use these data to estimate the life-cycle profiles. The outcomes are the time-use shares used in the model, specifically the share of weekly hours devoted to leisure, work in the market goods sector, work in the market services sector, home-produced services, and total market work. I treat leisure as the residual of a 168-hour weekly endowment. I also estimate a profile for log wages on the subsample of individuals employed full- or part-time. Because the weekend time-use module is collected only in odd years, I group ages into two-year bins and restrict the sample to ages 25 to 54 to match the model horizon. For each outcome I estimate a person fixed-effects regression of the variable on a full set of age-bin dummies, controlling for household size and the national unemployment rate, with standard errors clustered at the individual level. The specification is the

[4] Federal Reserve Bank of St. Louis, Consumer Price Index: Total for Germany [DEUCPIALLAINMEI], retrieved from FRED, https://fred.stlouisfed.org.

following, for each outcome y:

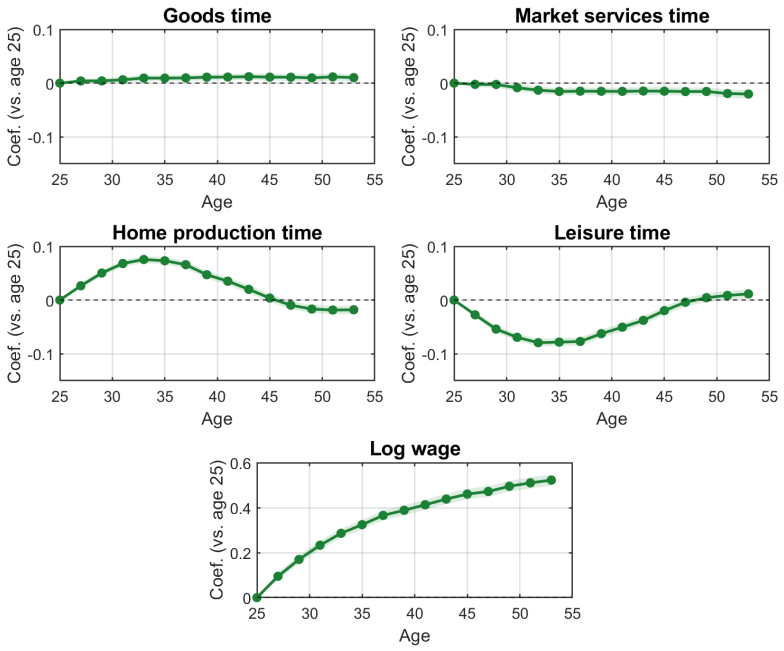
$$y_{i,t} = \alpha_i + \sum_k \beta_k \mathbf{1}[a_{i,t} \in \text{bin}_k] + \gamma\, N_{i,t} + \delta\, U_t + \varepsilon_{i,t}, \tag{1}$$

where α_i corresponds to the individual fixed effect and bin_k corresponds to the two-year age bins from 27 to 54, with the omitted bin (ages 25 to 26) serving as the reference, such that its coefficient is normalized to zero. $N_{i,t}$ is the household size, and U_t refers to the national unemployment rate. The person fixed effects absorb permanent unobserved heterogeneity, including cohort effects, and the unemployment rate captures aggregate business-cycle conditions. Thus, the age-bin coefficients, expressed relative to the youngest bin, trace out the life-cycle profile of each outcome net of cohort and business-cycle variation. I estimate the profiles separately for the pooled sample and for men and women, and report the resulting coefficients in Figure 1 and Figure 2, respectively. Both the estimated coefficients and their standard errors are carried to the calibration in Section 5, where each moment is weighted by the inverse of its estimated variance.

Figure 1 presents the life-cycle profiles of time use and log wages relative to the age 25 baseline. The estimates reveal a hump-shaped pattern in home production that directly mirrors the U-shaped pattern in leisure, with the respective peak and trough occurring between ages 33 and 35. Disaggregating the data by gender highlights the asymmetries in household specialization that occur over the life cycle. During child-rearing years, the increase in home production is almost twice as large for women as for men. At the same time, men increase their hours in market goods and experience steeper wage growth, while women's market goods hours remain flat. In the later stages of the life cycle (age 35 and older), home production declines sharply for both genders. For women, this reduction is accompanied by a rapid recovery in leisure, which returns to baseline levels by age 45. In contrast, after men's initial drop in leisure, their level does not recover given their high hours of market work. This reflects the specialization along the life-cycle of the couple documented by Mazzocco, Ruiz, and Yamaguchi (2013).

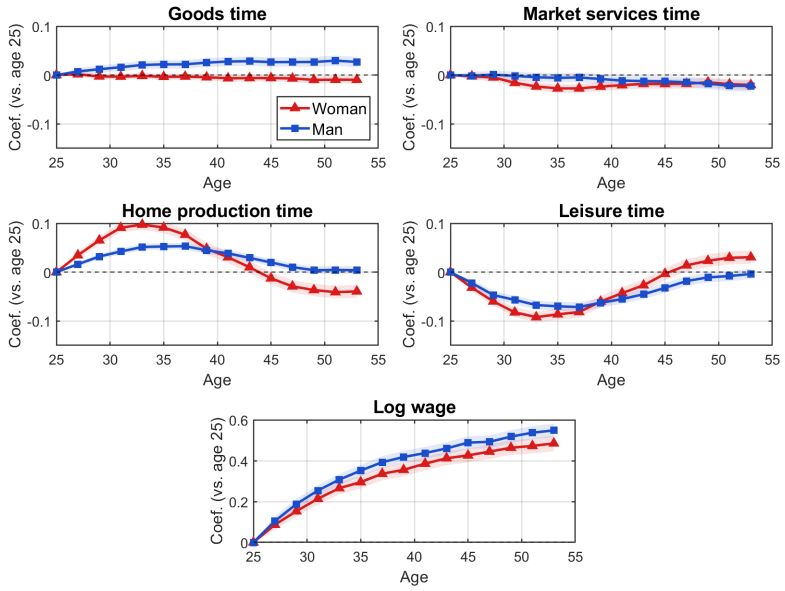
These profiles serve two purposes. First, they show that gender differences in time allocation are not constant but evolve systematically over the working life, which mo-

Figure 1: Life-cycle profiles of time use and wage growth.



Note: Profiles are estimated on the SOEP sample restricted to individuals aged 25–54. The sample includes only individuals living in couple households (with or without children). Log-wage profiles are estimated for employed individuals only.

Figure 2: Life-cycle profiles of time use and wage growth by gender.



Note: Profiles are estimated on the SOEP sample restricted to individuals aged 25–54. The sample includes only individuals living in couple households (with or without children). Log-wage profiles are estimated for employed individuals only.

tivates the life-cycle features of the model. Specifically, home production needs that vary with age, human capital accumulation that becomes harder with age, and household specialization that emerges gradually. Second, they provide empirical targets to discipline the model's parameters, as described in Section 5.

4 The Overlapping Generations Model

In this section, I extend the model of Duval-Hernández, Fang, and Ngai (2023) by considering an overlapping generations (OLG) collective model of the household, instead of the static and unitary model they propose. This allows me to capture two key mechanisms their study abstracts from, both of which the literature has shown to be relevant for labor supply decisions: dynamic incentives through human capital accumulation and intra-household bargaining [5]. First, considering a dynamic structure makes it possible to study not only the influence of the policy on current labor supply, but also its impact on future decisions through human capital accumulation: higher market hours today raise future wages and, in turn, increase the returns to future market work. Second, adopting a collective model allows me to account for the re-balancing of power within the household that takes place after the policy, given that this policy disproportionately affects the spouse specializing in home production by changing their outside option. In this way, the policy has an effect on household decisions through two distinct channels. First, it changes current and future incentives for work. Second, while it does not directly change the individual preferences of the spouses, it shifts their relative bargaining weights. Thus, the household's collective preferences will change, since now the household will place a different weight on the individual preferences of

[5]For example, Blundell, Costa Dias, Meghir, and Shaw (2016) show that the long-term effects of policy depend on the returns to experience, while Wallenius (2011) demonstrates that ignoring endogenous human capital accumulation through learning by doing severely biases the estimate of labor supply elasticities. Knowles (2012) argues that accounting for household-bargaining is essential in the analysis of policies that affect men and women differently.

the spouse specializing in home production [6].

I use the model to quantify the different mechanisms, evaluate the long-run effects of the policy, and perform counterfactual analyses. Specifically, the OLG model incorporates three features. First, home-production needs differ with age, being highest during midlife years and declining afterward.[7] Second, human capital accumulation becomes harder with age, such that the penalty from a career interruption depends on its timing, being especially severe around mid-career (Adda, Dustmann, and Stevens (2017)). Third, specialization within households evolves gradually over time, consistent with the patterns documented by Mazzocco, Ruiz, and Yamaguchi (2013).

The model is an OLG, multi-sector collective model with home production. The households consist of a woman and a man, denoted by the superscript $g \in \{w, m\}$. Age is discrete and indexed by a, with each period representing one year. I focus on individuals aged 25–54, therefore restricting attention to those who have generally completed their education but are not yet close to retirement. I interpret age 1 in the model to be age 25 in the data, thus focusing on $a = 1, \ldots, 30$. Each period a new household is born at age 25 and the household aged 54 dies, thus 30 cohorts coexist at any point in time. Since the population is constant, the stationary age distribution is uniform, and aggregates are unweighted sums over ages. Two market sectors produce market goods and market services, indexed by $j \in \{1, 2\}$. Home production delivers home services, denoted by subscript n, which are close substitutes for market services.

4.1 Individual Preferences and Constraints

Each spouse has preferences over their own total consumption C_a^g, and the joint household leisure L_a according to,

$$U^g(C_a^g, L_a) = \ln(C_a^g) + \theta^g \ln(L_a), \tag{2}$$

[6] This channel will matter only if the spouses' individual preferences differ. If they coincide exactly, shifts in bargaining weights leave the household's collective preferences unchanged.

[7] See Aguiar and Hurst (2016), Section 6, for empirical evidence on life-cycle profiles of time use.

where θ^g indicates the relative weight that spouses place on household leisure and varies by gender [8]. The total consumption C_a^g of each spouse is a composite of their own market good consumption $c_{1,a}^g$ and the household total services Q_a, while household total services Q_a are a composite of the household market services $c_{2,a}$ and home-produced services $c_{n,a}$,

$$C_a^g = \left[\omega(c_{1,a}^g)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)(Q_a)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \tag{3}$$

$$Q_a = \left[\gamma_a(c_{2,a})^{\frac{\sigma-1}{\sigma}} + (1-\gamma_a)(c_{n,a})^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}. \tag{4}$$

Notice that only the consumption of market goods has a gender superscript g, implying that market goods are treated as private, while market services, home-produced services, and leisure are treated as public within the household. [9] Here, ω measures the relative weight that the household assigns to market goods compared to total services. Similarly, γ_a is the relative weight that the household assigns to market services compared to home services, and depends on age [10]. This is included to capture the different

[8] I consider gender-specific weights for leisure to capture heterogeneous preferences over labor and leisure. Two remarks are in order. First, θ^g does not govern how much the household values the leisure time of spouse g. Since leisure is public, the division of leisure between spouses is determined by α_l, the weight on the woman's leisure in the production of L_a, together with the spouses' relative shadow wages. The parameters θ^w and θ^m instead govern the household's overall valuation of joint leisure. Second, because both spouses value the same L_a, allocations depend on these parameters only through their average weighted by the bargaining weights, $\mu\theta^w + (1-\mu)\theta^m$, where μ denotes the woman's Pareto weight introduced below. Any pair with the same weighted average generates identical allocations, so the two parameters are not separately identified from steady-state moments. I therefore set $\theta^m = \theta^w$ in the calibration, a normalization for a given bargaining weight. Gender-specific weights matter independently only when the bargaining weight moves: with $\theta^w > \theta^m$, for example, an increase in the woman's weight raises the household's effective taste for leisure.

[9] While not all goods are strictly private nor all services strictly public, this classification captures the idea that most home-produced services and their market substitutes are shared. For example, private consumption could include individual purchases such as clothing, whereas public services could include childcare or household cleaning, which benefit all household members. This approach is consistent with the common treatment of home production as public consumption in collective household models.

[10] Specifically, I consider a quadratic function for the weight, i.e. $\gamma_a = \underline{\gamma} + \left(\gamma - \underline{\gamma}\right)\left(\frac{a-a_\gamma}{a_\gamma}\right)^2$, where γ is the baseline weight at the start of working life and $\underline{\gamma}$ is the lowest possible weight, which is

home-production needs along the life cycle (for example, increasing during child-rearing years). The parameters ε and σ indicate the elasticity of substitution between goods and services, and between market services and home services, respectively. I assume that $\varepsilon < 1$ and $\sigma > 1$, implying that goods and services are complements to each other, while market and home services are imperfect substitutes [11]. The total household consumption of market goods is defined as the sum of the individual consumption, i.e. $c_{1,a} \equiv (c^w_{1,a} + c^m_{1,a})$. Home services are produced using total home-production hours $H_{n,a}$ as the only input, according to technology A_n,

$$c_{n,a} = A_n H_{n,a}. \tag{5}$$

Total home-production hours are a composite of the woman's and the man's home-production hours, $h^w_{n,a}$ and $h^m_{n,a}$, respectively,

$$H_{n,a} = \left[\alpha_n (h^w_{n,a})^{\frac{\eta-1}{\eta}} + (1-\alpha_n)(h^m_{n,a})^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}, \tag{6}$$

where α_n determines the woman's relative hours in home production and captures differences in spouses' productivity, preferences, or social norms. The parameter η refers to the elasticity of substitution of home-production hours between women and men. Similarly, total leisure L_a is a composite of the woman's and the man's leisure time, l^w_a and l^m_a, respectively,

$$L_a = \left[\alpha_l (l^w_a)^{\frac{\eta_l-1}{\eta_l}} + (1-\alpha_l)(l^m_a)^{\frac{\eta_l-1}{\eta_l}}\right]^{\frac{\eta_l}{\eta_l-1}}. \tag{7}$$

α_l refers to the weight of the woman in the total leisure of the household. I assume that $\eta_l < 1$, implying that spouses' leisure times are complements to each other [12]. Each

realized at age a_γ. Since γ_a is the weight on market services, its trough corresponds to the peak of home-production needs.

[11] These assumptions are in line with estimates found in the literature. For estimates of the elasticity of substitution between market and home services, see the survey from Aguiar, Hurst, and Karabarbounis (2012). For estimates of the elasticity of substitution between goods and services, see Herrendorf, Rogerson, and Valentinyi (2013) and Moro, Moslehi, and Tanaka (2017).

[12] I adopt the same assumption as Duval-Hernández, Fang, and Ngai (2023), who motivate it with empirical evidence from Goux, Maurin, and Petrongolo (2014) and references therein.

spouse is endowed with one unit of time per period, which they divide among leisure l_a^g, home production $h_{n,a}^g$, and market work h_a^g,

$$1 = l_a^g + h_{n,a}^g + h_a^g. \tag{8}$$

Spouses are endowed with initial human capital $z_{a=1}^w, z_{a=1}^m$. They accumulate human capital only when working in the market [13], according to the following law of motion

$$z_{a+1}^g = (1-\delta)z_a^g + B_a z_a^g (h_a^g)^\nu, \tag{9}$$

which states that next period's human capital depends on today's human capital net of depreciation — where δ is the depreciation rate of human capital — plus the accumulation from current working hours, with ν governing the curvature of the human capital technology. $B_a = Be^{-da}$ acts as a scaling factor, with d representing the rate at which B_a declines with age [14]. Thus, the human capital forgone by missing market work depends on the career stage, with persistent consequences for future wages and labor supply. Notice that by incorporating human capital accumulation, the model endogenously generates gradual specialization. Small initial differences in spouses' labor supply decisions lead to differences in human capital and wages. These differences amplify over time, generating a gradual process of specialization, with one spouse focusing more on market work and the other on home production.

4.2 Household Decision Process

I assume that the household decision process follows the collective model with limited commitment framework [15]. This framework departs from the unitary model, which

[13] Human capital through learning-by-doing occurs exclusively in market work, not in home production. This assumption is consistent with empirical evidence on the negative effects of career interruptions on wages. See Albrecht, Edin, Sundström, and Vroman (1999) and references therein.

[14] This specification follows Wallenius (2011), who uses it to capture depreciation, learning, complementarity between current human capital and hours worked, and the possibility that learning becomes harder with age.

[15] The limited commitment assumption is supported by empirical evidence on household decision-making. Mazzocco (2007) and Lise and Yamada (2018) reject full commitment in favor of weaker

treats the household as a single decision-making agent, by recognizing that households typically consist of multiple members with distinct preferences. Following the collective approach (Chiappori (1988), Chiappori (1992), Apps and Rees (1988)), intra-household allocations are assumed to be Pareto efficient, without requiring an explicit specification of the bargaining process. Under limited commitment, however, spouses cannot fully commit to future allocations. Consequently, at each point in time, household decisions must satisfy individual participation constraints, ensuring that both spouses are at least as well off within the household as they would be with their outside option. Consistent with the literature, I assume that the outside option corresponds to the value of being single [16].

Household maximization The household chooses how much to consume in each sector $(c^w_{1,a}, c^m_{1,a}, c_{2,a})$ and how to allocate their time $(h^w_a, h^m_a, h^w_{n,a}, h^m_{n,a}, l^w_a, l^m_a)$ to maximize discounted lifetime utility of the household, which is defined as the weighted sum of the spouses' individual utilities,

$$\max \sum_{a=1}^{30} \beta^{a-1} \left[\mu_a U^w(C^w_a, L_a) + (1-\mu_a) U^m(C^m_a, L_a)\right], \tag{10}$$

where $\beta \in (0,1)$ is the discount factor. μ_a and $(1-\mu_a)$ represent the weights assigned to the woman's and the man's preferences, respectively. In empirical settings, bargaining weights may depend on spouses' wages and earnings, as well as a set of distributional factors that do not affect preferences and prices, such as social norms, ownership of assets, and divorce legislation [17]. In this model, however, the bargaining weights are not specified as an explicit function of such variables. Instead, they adjust endogenously whenever a spouse's participation constraint binds. This adjustment process is discussed in detail when I introduce the participation constraints below.

frameworks, while Theloudis, Velilla, Chiappori, Giménez-Nadal, and Molina (2025) explicitly test and find support for limited commitment.

[16] See, for example Mazzocco (2007) and Lise and Yamada (2018).

[17] See Campaña, Giménez-Nadal, Molina, and Velilla (2024) for a recent empirical analysis of the determinants of intrahousehold bargaining power for different European countries.

Budget constraint Household decisions are subject to their following per-period budget constraint,

$$p_1(1+\tau_1)\left(c_{1,a}^w + c_{1,a}^m\right) + p_2(1+\tau_2)\, c_{2,a} = (1-\tau_h)\left(w^w h_a^w z_a^w + w^m h_a^m z_a^m\right) + T_a, \quad (11)$$

where p_1 and p_2 refer to the price of market goods and market services, respectively, and τ_1, τ_2 and τ_h correspond to the taxes levied by the government on market goods, market services, and labor income, respectively. The revenue from these taxes is returned to the household as a lump-sum transfer T_a. For simplicity, the model assumes there are no savings, i.e. households consume their entire after-tax income each period.

Participation and renegotiation Since I adopt a limited commitment framework, the household's problem is also subject to participation constraints, ensuring that each spouse prefers to remain in the household rather than exercising their outside option. Thus, taking $c_{1,a}^{w*}, c_{1,a}^{m*}, c_{2,a}^{*}, h_{n,a}^{w*}, h_{n,a}^{m*}, l_a^{w*}, l_a^{m*}$ as the solution of the household problem, the value of remaining in the couple for spouse g, given current bargaining weights μ_a and $(1-\mu_a)$, is defined recursively as

$$V_a^{g,M}(z_a^g, z_a^{-g}, \mu_a) = U^g(C_a^{g*}, L_a^*) + \beta V_{a+1}^{g,M}(z_{a+1}^g, z_{a+1}^{-g}, \mu_{a+1}), \qquad \text{for } g \in \{w, m\}, \quad (12)$$

where μ_{a+1} reflects any renegotiation of the bargaining weight, and human capital evolves according to its law of motion. The outside option of each spouse is the value of leaving the household and remaining single for the rest of the life cycle, $V_a^{g,S}(z_a^g)$, defined in Appendix A.1. A spouse carries the human capital z_a^g accumulated within the marriage into singlehood. The participation constraint then requires

$$V_a^{g,M}(z_a^g, z_a^{-g}, \mu_a) \geq V_a^{g,S}(z_a^g), \qquad \text{for } g \in \{w, m\}. \quad (13)$$

Given the bargaining weight from the previous period μ_{a-1}, the household checks the two participation constraints (13). If both hold, the weight is unchanged, $\mu_a = \mu_{a-1}$. If one spouse's constraint is violated, the household renegotiates: bargaining weights adjust in favor of the constrained spouse just enough to restore their participation. The weights consistent with both constraints form an interval $[\underline{\mu}_a, \overline{\mu}_a]$, which is non-empty

whenever the surplus from remaining together is positive for both spouses. In the calibrated stationary equilibrium this surplus is positive at every age, so the interval is never empty, the representative couple stays intact over the life cycle, and dissolution operates only as an off-equilibrium threat that disciplines the path of μ_a [18]. Notice that past labor supply decisions affect current human capital, which determines each spouse's outside option and, through the participation constraint, the bargaining weights. Thus, previous-period work influences current decisions both via higher returns to human capital and via changes in bargaining weights (see Basu (2006)).

4.3 Firms

For the firms, their problem repeats every period, since there are no dynamics connecting their decisions. Thus, I eliminate the a subscript. Market goods and market services are produced each by one representative firm which uses labor as its only input, according to the time-invariant technology A_j in the following way,

$$Y_j = A_j H_j, \tag{14}$$

Labor can move freely across sectors, such that wages are equalized across sectors. The total hours H_j used by the firms in their production function corresponds to a composite of the women's and the men's hours.

$$H_j = \left[\alpha_j (H_j^w)^{\frac{\eta-1}{\eta}} + (1-\alpha_j)(H_j^m)^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}. \tag{15}$$

The woman's intensity in the production function, represented by α_j, captures the different factors affecting such intensities in production, for example, differences in productivity between men and women or social norms. I allow for the intensities to differ by sector, and assume that the woman's intensity in the production of market goods is lower than the intensity in the production of market services, $\alpha_1 < \alpha_2$, implying that the woman has a comparative advantage in the production of services [19]. The parameter η

[18] Singles therefore do not enter the stationary aggregates.

[19] This assumption follows Ngai and Petrongolo (2017) and is supported by evidence on gender differences in interpersonal and communication skills, which remain important in service tasks (Borghans, Duckworth, Heckman, and Weel (2008)).

refers to the elasticity of substitution between women and men. H_j^g denotes the total effective units of labor from gender $g \in \{w, m\}$ demanded in sector j. Since wages equalize across sectors due to free mobility of labor, the household side determines only the aggregate effective supply $\sum_{a=1}^{30} z_a^g h_a^g$ of each gender. Therefore, the sectoral allocation of that labor is pinned down by the firms' first-order conditions together with the labor-market-clearing conditions below. Firm profits are

$$\pi_j = p_j Y_j - w^w H_j^w - w^m H_j^m, \qquad \text{for } j \in \{1, 2\}. \tag{16}$$

4.4 Government

The government levies consumption taxes τ_1, τ_2 and a labor income tax τ_h, and redistributes revenues as transfers T_a, satisfying:

$$\{T_a = p_1\tau_1(c_{1,a}^w + c_{1,a}^m) + p_2\tau_2 c_{2,a} + \tau_h\left(w^w h_a^w z_a^w + w^m h_a^m z_a^m\right)\}_{a=1}^{a=30}. \tag{17}$$

4.5 Equilibrium

A stationary equilibrium consists of:

(i) Stationary prices: $\{w^w, w^m, p_1, p_2\}$, with the numeraire $p_1 = 1$.

(ii) Lump-sum transfers: $\{T_a\}_{a=1}^{a=30}$.

(iii) Firm labor demand: $\{H_1^{w,D}, H_1^{m,D}, H_2^{w,D}, H_2^{m,D}\}$.

(iv) Representative cohort life-cycle allocations $\{c_{1,a}^w, c_{1,a}^m, c_{2,a}, c_{n,a}, h_a^w, h_a^m, h_{n,a}^w, h_{n,a}^m, l_a^w, l_a^m\}_{a=1}^{a=30}$, and life-cycle states $\{z_a^w, z_a^m, \mu_a\}_{a=1}^{a=30}$.

such that

(i) Firms optimize: taking prices as given, each sector $j \in \{1, 2\}$ chooses effective units of gender labor demand $H_j^{w,D}, H_j^{m,D}$ to maximize its profits (16).

(ii) Households optimize: Taking prices and the transfer T_a as given, the representative household solves the limited-commitment collective problem, maximizing

(10) subject to the per-period budget constraint (11), the time constraint (8), the home-production technology (5), the human-capital technology (9), and the spouses' participation constraints.

(iii) Markets clear:

1. Goods market $\quad Y_1 = \sum_{a=1}^{30}(c_{1,a}^w + c_{1,a}^m)$.
2. Services market $\quad Y_2 = \sum_{a=1}^{30} c_{2,a}$.
3. Labor market $\quad \{H^{g,D} = H_1^{g,D} + H_2^{g,D} = \sum_{a=1}^{30} z_a^g h_a^g\}^{g=\{w,m\}}$.

(iv) Government budget constraint is satisfied for all a:
$\{T_a = p_1\tau_1(c_{1,a}^w + c_{1,a}^m) + p_2\tau_2 c_{2,a} + \tau_h\left(w^w h_a^w z_a^w + w^m h_a^m z_a^m\right)\}_{a=1}^{a=30}$.

(v) Laws of motion: human capital follows $z_{a+1}^g = (1-\delta)z_a^g + B_a z_a^g (h_a^g)^\nu$. The bargaining weight μ_a evolves endogenously as the participation constraints bind. Initial states $z_{a=1}^w$, $z_{a=1}^m$, and $\mu_{a=1}$ are given.

I derive the equilibrium and explain the algorithm used to solve the model in Appendices A.2 and A.3, respectively.

5 Preliminary Calibration

I calibrate the model to the German economy at an annual frequency, with each period corresponding to one year and individuals followed from age 25 to age 54. The strategy follows Duval-Hernández, Fang, and Ngai (2023) and French (2005). Parameters are split into two groups. The first group is set externally, using standard values from the literature. Table 1 reports the parameter values and their sources. The second group is calibrated internally so that the model reproduces a set of empirical targets. Table 2 reports these calibrated parameters, their values, and the moments they target. I normalize the price of the goods sector to one, $p_1 = 1$, and assume that couples enter at age 25 with the same initial human capital, $z_{a=1}^w = z_{a=1}^m$, and equal bargaining weights, $\mu_{a=1} = 0.5$.

5.1 Externally Calibrated Parameters

The externally set parameters are the discount factor, the depreciation rate of human capital, the elasticities of substitution, and the tax rates. For the elasticities I follow the literature: the elasticity of substitution between goods and services, ε, is taken from Cravino and Sotelo (2019); the elasticity between market and home services, σ, the elasticity between women and men labor in production, η, and the corresponding elasticity in leisure, η_l, from Duval-Hernández, Fang, and Ngai (2023). Women and men labor are imperfect substitutes within each sector, while they are complements in total household leisure ($\eta > 1$ and $\eta_l < 1$). The consumption tax on goods, τ_1, and the labor income tax, τ_h, are likewise set to the German values in McDaniel (2023). In the baseline, the consumption tax on substitutable market services is also taken from McDaniel (2023), $\tau_2 = 0.19$. The policy experiment in Section 6 lowers τ_2. I take the depreciation rate of human capital from Wallenius (2011), while the discount factor β is set to a standard annual value, as in Wallenius (2011).

Table 1: Externally calibrated parameters

	Values	**Description**	**Source**
β	0.96	Discount factor	Wallenius (2011)
δ	0.03	Depreciation rate of human capital	Wallenius (2011)
η	3	Elast. of subst. b/w genders in work	Duval-Hernández et al. (2023)
η_l	0.2	Elast. of subst. b/w genders in leisure	Duval-Hernández et al. (2023)
ε	0.2	Elast. of subst. b/w goods and services	Cravino and Sotelo (2019)
σ	1.9	Elast. of subst. b/w market and home services	Duval-Hernández et al. (2023)
τ_1	0.19	Consumption tax for sector 1	McDaniel (2023)
τ_2	0.19	Consumption tax for sector 2	McDaniel (2023)
τ_h	0.38	Labor income tax	McDaniel (2023)

5.2 Internally Calibrated Parameters

The internal parameters divide into two blocks. The first governs the level of time use and its division across sectors and genders. The second governs how time use and wages evolve over the life cycle.

Cross-sectional block The first block comprises the CES weights on the woman's hours in each activity, α_1, α_2, α_n, and α_l; the sectoral productivities A_1 and A_2; and the leisure preferences θ^w and θ^m. Following Duval-Hernández, Fang, and Ngai (2023), the gender weights are pinned down by the observed gender intensity of hours in each activity, which is what gives women a comparative advantage in market services and in home production. The productivities are identified by the sectoral composition of hours: A_1 by the ratio of hours in market goods to hours in market services, and A_2 by the ratio of hours in market to home services. Because the CES weights on goods, ω, and on market services, γ, cannot be separately identified from the relative productivities, I set them and let the productivities match the two sectoral ratios, with the home productivity normalized to $A_n = 1$. Since both spouses value the same joint leisure, household allocations depend on the leisure preferences only through the bargaining-weighted average $\mu\theta^w + (1-\mu)\theta^m$, so the two parameters are not separately identified (see footnote 8). I therefore impose $\theta^m = \theta^w$ and identify the common value by the ratio of leisure to total market hours.

Life-cycle block The second block comprises the parameters of the human capital technology, namely the scale of accumulation B, the rate at which the ability to accumulate declines with age d, and the curvature of accumulation in hours ν, together with the parameters of the age profile of home-production needs, the baseline weight γ, the trough level $\underline{\gamma}$, and the trough age a_γ. Following French (2005), I discipline the first group with three moments of the life-cycle wage profiles estimated in Section 3, while the parameters of the home-production profile are set directly.

The scale of accumulation, B, is identified by the cumulative wage growth over the working life, averaged across genders. The rate at which the ability to accumulate

declines with age, d, is identified by the concavity of the wage profile, measured as the ratio of late-career (ages 45 to 54) to early-career (ages 25 to 35) wage growth, again pooled across genders. This same decline makes the cost of a career interruption largest early in the career, consistent with Adda, Dustmann, and Stevens (2017). The curvature parameter, ν, which governs the elasticity of human capital accumulation with respect to hours worked, is identified by the gender gap in wage growth. Since identification requires variation in hours, I exploit the difference in market hours between spouses, who share the same B and the same d. Given that difference, the gender gap in cumulative wage growth pins down ν. The three parameters are thus identified by three moments: pooled cumulative wage growth, the pooled ratio of late- to early-career growth, and the gender gap in cumulative growth.

The age profile of home-production needs is a symmetric quadratic in age, which attains its trough $\underline{\gamma}$ at age a_γ and equals the baseline weight γ at the start of working life. Since γ_a is the weight on market services, its trough corresponds to the peak of home-production needs. The profile is a household-level object rather than a gender-specific one, and I set its parameters directly, with $\gamma = 0.535$, $\underline{\gamma} = 0.465$, and $a_\gamma = 15$. The larger hump in women's home production does not come from this profile, but from the gender weight in home production together with the specialization that the model generates endogenously as spouses' human capital diverges.

Table 2 reports the calibrated values, the description of the parameters, and the moments each parameter targets.

Table 2: Internally calibrated parameters

	Values	Description	Data Target
Group 1: cross-sectional moments			
θ_w	1.6	Leisure preference (woman)	Ratio leisure to market goods hours (woman)
θ_m	1.6	Leisure preference (man)	Ratio leisure to market goods hours (man)
α_1	0.552	CES weight on woman's labor in sector 1	Gender ratio of hours in market goods
α_2	0.634	CES weight on woman's labor in sector 2	Gender ratio of hours in market services
α_n	0.670	CES weight on woman's home prod. time	Gender ratio of home hours
α_l	0.520	CES weight on woman's leisure	Gender ratio of leisure
A_1	5	Productivity in sector 1	Ratio market goods to market services
A_2	0.368	Productivity in sector 2	Ratio market to home services
A_n	1	Productivity in home sector (normalized)	
ω	0.5	Weight of sector 1's goods (set)	
Group 2: life-cycle profiles			
B	0.12	Human capital accumulation parameter	Slope of life-cycle wage profile
d	0.022	Rate of decay of HC accumulation	Decline of wage growth with age
ν	0.3	Human capital elasticity $(h^g)^\nu$	Wage returns to hours
γ	0.535	Baseline weight of market services (set)	
a_γ	15	Age of minimum market services weight (set)	
$\underline{\gamma}$	0.465	Minimum market services weight (set)	

5.3 Model Fit

The model reproduces the targeted cross-sectional ratios closely (Table 3). The four gender ratios of hours, in market goods, market services, home services, and leisure, are matched to within about eight percent of their empirical counterparts, at 0.334 against 0.345, 0.929 against 1.007, 1.673 against 1.772, and 0.913 against 0.976, respectively. The three sectoral hours ratios are essentially on target: the goods-to-services ratio is 0.426 against 0.426 in the data, the market-to-home services ratio is 0.977 against 0.944, and the leisure-to-market-goods ratio is 2.088 against 2.092.

The targeted life-cycle wage moments are matched with mixed success. The model

Table 3: Data and Model Moments

Symbol	Data	Model	Description
Targeted moments: cross-sectional moments			
H_1^w/H_1^m	0.345	0.334	Gender ratio in market goods
H_2^w/H_2^m	1.007	0.929	Gender ratio in market services
H_n^w/H_n^m	1.772	1.673	Gender ratio in home services
L^w/L^m	0.976	0.913	Gender ratio in leisure
H_1/H_2	0.426	0.426	Hours ratio b/w market goods and services
H_2/H_n	0.944	0.977	Hours ratio b/w market and home services
H_l/H_1	2.092	2.088	Hours ratio b/w leisure and market goods
Targeted moments: life-cycle profiles			
$log(w_{30}/w_1)$	0.517	0.473	Wage growth over the life-cycle
$\frac{log(w_{30}/w_{21})}{log(w_{10}/w_1)}$	0.207	0.211	Ratio of late- to early-career wage growth
$log(w_{30}^w/w_1^w) - log(w_{30}^m/w_1^m)$	-0.064	-0.135	Gender gap of wage growth

reproduces the level of wage growth over the working life reasonably well, at 0.473 log points against 0.517 in the data, and it matches the deceleration of that growth almost exactly, with a ratio of late- to early-career wage growth of 0.211 against 0.207. It does less well on the gender dimension: the modeled gender gap in wage growth, at -0.135 log points, is roughly twice its empirical value of -0.064, so the model overstates how much faster men's wages grow than women's over the life cycle.

The life-cycle profiles in Figures 3 and 4 locate the fit more precisely. The model captures the relative magnitudes of time use and wages by gender and tracks the wage profiles well, matching men's wage growth almost exactly while understating women's, which is consistent with the overstated gender gap. Three features remain poorly reproduced. First, the model does not generate the rising life-cycle profile of market goods hours: in the pooled data goods time increases with age, whereas the model implies a flat-to-U-shaped profile of the opposite sign, and it misses the pronounced

rise in men's goods time. This U-shape may stem from the age-varying weight in the CES aggregator for services, which shifts the desired mix of market and home services over the life cycle and feeds back into the demand for goods. Second, the hump in home-production time is too flat and peaks too late, reaching about 0.03 to 0.04 in the early forties against a data peak near 0.08 to 0.10 in the early thirties, and the model does not capture the sharp decline in home hours after the mid-thirties, particularly for women. Third, the model overstates the mid-life decline in women's market services time while men's modeled profile recovers toward the end of the working life, whereas both decline in the data.

The remaining mismatches are dynamic rather than cross-sectional. The model reproduces the relative size of the sectors and the gender ratios in time use, but it does not capture the full shape of the life-cycle profiles, in particular the amplitude and timing of the home-production hump and the rising profile of market goods hours. Because the policy effects accumulate over the life cycle, the provisional magnitudes reported below should be read with these profile mismatches in mind. Refining the fit of these profiles, together with the gender wage gap, is the focus of ongoing work.

Figure 3: Model fit of life-cycle profiles of time use and wage growth.

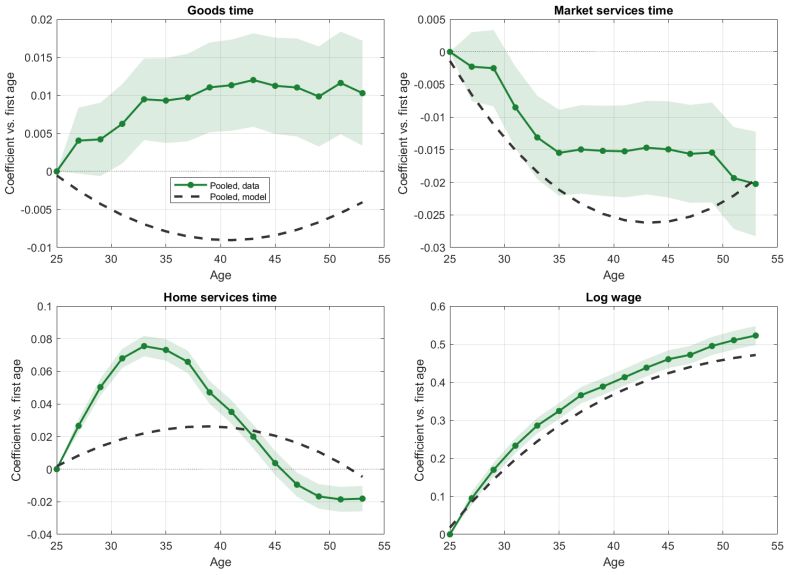
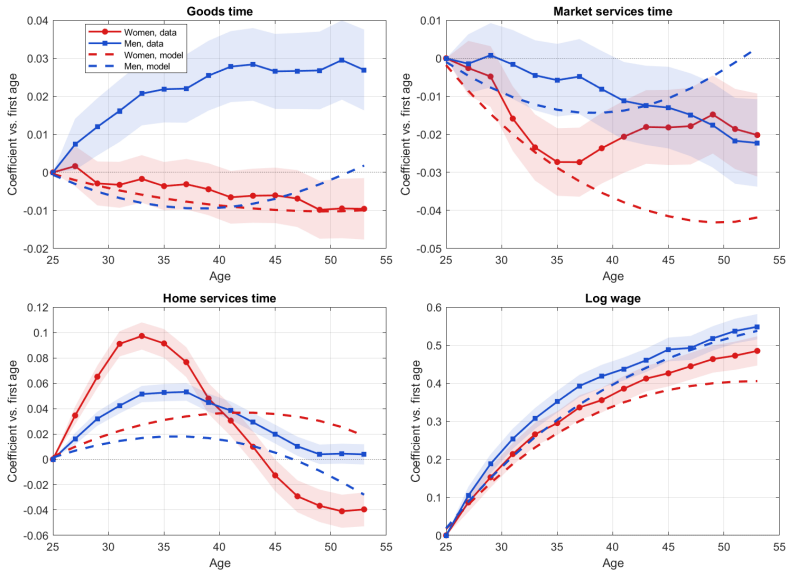


Figure 4: Model fit of life-cycle profiles of time use and wage growth by gender.



6 Quantitative results

Starting from the calibrated baseline economy, I study a counterfactual reform that lowers the consumption tax on substitutable market services, τ_2, by 25%, from 0.19 to 0.1425. All other parameters and tax rates are held at their calibrated values. I recompute the equilibrium and compare the resulting allocations, prices, and gender gaps in hours and wages to the baseline. Because the model is solved in general equilibrium, the experiment captures not only the direct effect of the lower tax but also the adjustment of the wages w^w and w^m, the price of market services p_2, the life-cycle paths of human capital, and the bargaining weights within the household.

The reform is hypothetical, but it lies within the range of policies already in place in Germany and across the European Union, and it follows a direction advocated by the European Commission as a means of raising female labor supply and formalizing domestic work (Manoudi, Weber, Scott, and Hawley Woodall (2018)). Lowering the tax on substitutable market services reduces the wedge between purchasing these services in the market and producing them at home. Germany already reduces the cost of household services through the tax credit for household-related services (*Steuerermäßigung*

für haushaltsnahe Dienstleistungen), a 20 percent credit on the labor cost of such services up to an annual maximum of EUR 4,000. Several other countries operate larger schemes: France (*CESU*, 50 percent), Belgium (*titres-services*), Sweden (*RUT-avdrag*, 50 percent), and Finland (*kotitalousvähennys*, 40 percent), studied in Section 2, with reductions in the cost of these services at the maximum subsidy of roughly 40 to 63 percent (OECD (2021a)). Reducing τ_2 by 25 percent lowers the consumer price of substitutable market services, incentivizing households to substitute home-produced services with market services. However, because the model lacks a distinction between substitutable and non-substitutable services, any simulated tax reduction necessarily applies to the entire services sector. This mechanical limitation means the policy's effects in the model reflect a broader subsidy than the targeted real-world schemes discussed above, which must be accounted for when interpreting the quantitative results.

In the remainder of this section I report the effects of the reform on the gender gaps in market hours and wages, and I trace how these effects build up over the life cycle. In ongoing work, I plan to decompose the total response into the contributions of human capital accumulation, intra-household bargaining, and the general equilibrium adjustment of prices, and examine how the effects depend on the age at which the reform is introduced.

The reform shifts the composition of output toward services. As the consumption tax on substitutable market services falls, households substitute market services for home production: service output rises by about 8 percent while goods output rises by only about 1 percent, so production tilts toward services even though both sectors expand. The producer price of services is almost unchanged, falling by 0.04 percent, so the tax cut passes through almost entirely to consumers, as a fall of about 4 percent in the consumer price of these services, which drives the quantity response.

The reform raises women's market work and narrows the gender gap in market hours. Women's aggregate market hours increase by 5.8 percent, against 4.1 percent for men, because women hold a comparative advantage in services and perform most home production, so the marketization of home services frees proportionally more of

Table 4: Macroeconomic Effects of the Reform

	Baseline	**Reform**	**% Change**
Group 1: Prices and Wages			
w^w	3.421	3.418	-0.110
w^m	1.873	1.874	0.062
p_2	14.365	14.359	-0.043
Gender wage ratio w^w/w^m	1.827	1.824	-0.171
Group 2: Time Allocation			
Market hours, women	2.965	3.137	5.798
Market hours, men	4.291	4.466	4.082
Market hours gap (w/m)	0.691	0.702	1.648
Home hours, women	7.934	7.797	-1.736
Home hours, men	4.778	4.652	-2.622
Home hours gap (w/m)	1.661	1.676	0.909
Group 3: Production Aggregates			
Output, goods Y_1	14.218	14.415	1.372
Output, services Y_2	2.459	2.658	8.096

their time for market work. Home production falls for both spouses, by 1.7 percent for women and 2.6 percent for men. The ratio of women's to men's market hours rises from 0.69 to 0.70, a narrowing of about 1.6 percent. The home-hours gap moves in the opposite direction, widening marginally by about 0.9 percent, because men reduce their home hours by proportionally more than women.

Figure 5: Time allocations and wages. Baseline vs. Reform

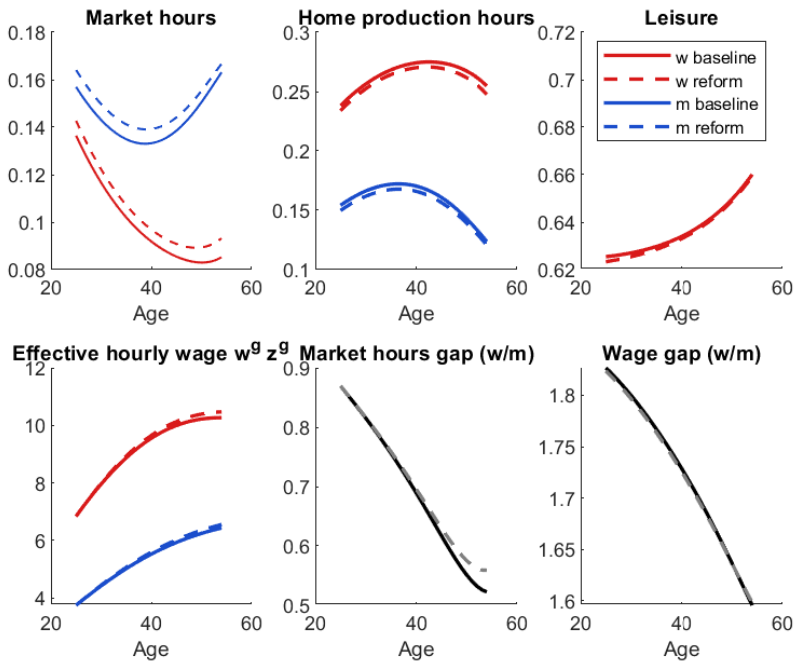


Figure 5 shows that these effects accumulate over the life cycle: in both economies women's market hours fall and their home production rises with age as specialization deepens, but the reform shifts women's market-hours profile upward throughout, partially offsetting the life-cycle widening of the gap.

The effect on wages is small. Women's wage per efficiency unit falls by 0.11 percent and men's rises by 0.06 percent, so the gender wage ratio w^w/w^m falls by about 0.17 percent. The effective hourly wage profiles are almost unchanged by the reform, so the life-cycle widening of the wage gap, which here reflects the divergence in human capital between spouses as men accumulate more market experience, is essentially unaffected. In this version of the model the reform therefore operates almost entirely on

the allocation of hours rather than on wages.

7 Conclusion

This paper analyzes how a policy that lowers taxes on market substitutes for home production shapes female labor supply and the gender gap in market hours over the life cycle. By developing and calibrating a multi-sector, overlapping generations collective model of the household for Germany, I extend the existing literature in two directions: incorporating life-cycle dynamics driven by human capital accumulation, and allowing for the endogenous renegotiation of bargaining power between spouses. The framework captures how a targeted fiscal policy reallocates time within the household and how these effects accumulate over the working life, margins that are absent in static or unitary settings.

The quantitative results show that lowering the consumption tax on substitutable market services reallocates activity toward the market. As the relative price of these services falls, households substitute market services for home production, and because women perform most home production and hold a comparative advantage in services, they raise their market hours by more than men. This narrows the gender gap in market hours, and the life-cycle structure shows that the increase accumulates over the working life, partially offsetting the specialization that widens the gap with age. In this version of the model the response operates almost entirely through the allocation of hours: wages per efficiency unit and the life-cycle wage profiles are essentially unchanged, so the reform leaves the gender wage gap largely unaffected.

For policy design, these results suggest that lowering taxes on home-production substitutes can shift time use toward the market and narrow the gender gap in hours, while in general equilibrium the expansion of the market service sector absorbs the additional female labor supply. However, the magnitudes are modest, and because the calibration is still preliminary, they are best interpreted as provisional.

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A Appendix

A.1 Single's Problem

A spouse who leaves the household solves a single-person version of the problem, taking equilibrium prices $\{p_1, p_2\}$ and wages $\{w^w, w^m\}$ as given and bringing the human capital z_a^g accumulated within the marriage. For an individual of gender $g \in \{w, m\}$, the value of being single is

$$V_a^{g,S}(z_a^g) = \max_{\{c_{1,a}^g, c_{2,a}^g, h_{n,a}^g, l_a^g\}} \left\{ \ln C_a^{g,S} + \theta^g \ln l_a^g + \beta\, V_{a+1}^{g,S}(z_{a+1}^g) \right\}, \qquad V_{31}^{g,S}(\cdot) = 0, \tag{18}$$

subject to

$$C_a^{g,S} = \left[\omega (c_{1,a}^g)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)(Q_a^{g,S})^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \tag{19}$$

$$Q_a^{g,S} = \left[\gamma_a (c_{2,a}^g)^{\frac{\sigma-1}{\sigma}} + (1-\gamma_a)(c_{n,a}^{g,S})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{20}$$

$$c_{n,a}^{g,S} = A_n\, h_{n,a}^g, \tag{21}$$

$$p_1(1+\tau_1)\, c_{1,a}^g + p_2(1+\tau_2)\, c_{2,a}^g = (1-\tau_h)\, w^g z_a^g h_a^g + T_a^S, \tag{22}$$

$$1 = l_a^g + h_{n,a}^g + h_a^g, \tag{23}$$

$$z_{a+1}^g = (1-\delta) z_a^g + B_a z_a^g (h_a^g)^\nu. \tag{24}$$

where T_a^S is the lump-sum transfer from the government to the single household. Relative to the couple's problem, the two CES aggregators over spouses collapse: leisure is the individual's own leisure l_a^g, thus the aggregator (7) and the complementarity η_l play no role. And similarly home services are produced from own hours alone, thus the aggregator (6) and the weight α_n do not appear [20]. Market and home services are no longer shared across two members, and there is no bargaining weight. The gap between $V_a^{g,M}$ and $V_a^{g,S}$ therefore reflects the value a spouse derives from shared public consumption, complementarity in joint leisure, and specialization within the couple.

[20] I assume a linear home-production technology for singles rather than the solo limit of (6) ($A_n\, \alpha_n^{\eta/(\eta-1)}$ for a woman and $A_n(1-\alpha_n)^{\eta/(\eta-1)}$ for a man) to avoid imposing a mechanical productivity penalty on single households driven purely by the couple's CES share and elasticity parameters.

A.2 Derivation of Stationary Equilibrium

Household Problem Household value function

$$V_a(z_a^w, z_a^m, \mu_a) = \max_{\{c_{1,a}^g, c_{2,a}, h_{n,a}^g, l_a^g\}^{g=\{w,m\}}} \left\{ U(c_{1,a}^w, c_{1,a}^m, c_{2,a}, h_{n,a}^w, h_{n,a}^m, l_a^w, l_a^m) + \beta V_{a+1}(z_{a+1}^w, z_{a+1}^m, \mu_{a+1}) \right\}, \tag{25}$$

where $U(c_{1,a}^w, c_{1,a}^m, c_{2,a}, h_{n,a}^w, h_{n,a}^m, l_a^w, l_a^m) = [\mu_a U^w(C_a^w, L_a) + (1-\mu_a) U^m(C_a^m, L_a)]$. And subject to:

$$C_a^g = \left[\omega (c_{1,a}^g)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)(Q_a)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \text{for } g = \{w, m\}, \tag{26}$$

$$Q_a = \left[\gamma_a (c_{2,a})^{\frac{\sigma-1}{\sigma}} + (1-\gamma_a)(c_{n,a})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{27}$$

$$c_{n,a} = A_n \left[\alpha_n (h_{n,a}^w)^{\frac{\eta-1}{\eta}} + (1-\alpha_n)(h_{n,a}^m)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \tag{28}$$

$$L_a = \left[\alpha_l (l_a^w)^{\frac{\eta_l-1}{\eta_l}} + (1-\alpha_l)(l_a^m)^{\frac{\eta_l-1}{\eta_l}} \right]^{\frac{\eta_l}{\eta_l-1}}, \tag{29}$$

$$p_1(1+\tau_1)(c_{1,a}^w + c_{1,a}^m) + p_2(1+\tau_2)c_{2,a} =$$

$$(1-\tau_h) \left[w^w z_a^w \underbrace{(1 - h_{n,a}^w - l_a^w)}_{=h_a^w} + w^m z_a^m \underbrace{(1 - h_{n,a}^m - l_a^m)}_{=h_a^m} \right] + T_a, \tag{30}$$

$$z_{a+1}^g = (1-\delta) z_a^g + B_a z_a^g (h_a^g)^\nu, \qquad \text{for } g = \{w, m\}, \tag{31}$$

$$h_a^g \geq 0, \qquad \text{for } g = \{w, m\}, \tag{32}$$

where $B_a = Be^{-da}$. Taking first order conditions:

$$\frac{\partial V_a(\cdot)}{\partial c_1^w}: \quad \mu_a \omega \frac{1}{C_a^w}\left(\frac{C_a^w}{c_{1,a}^w}\right)^{\frac{1}{\varepsilon}} - \lambda_a p_1(1+\tau_1) = 0, \tag{33}$$

$$\frac{\partial V_a(\cdot)}{\partial c_1^m}: \quad (1-\mu_a)\omega \frac{1}{C_a^m}\left(\frac{C_a^m}{c_{1,a}^m}\right)^{\frac{1}{\varepsilon}} - \lambda_a p_1(1+\tau_1) = 0, \tag{34}$$

$$\frac{\partial V_a(\cdot)}{\partial c_2}: \quad (1-\omega)\gamma_a \left(\frac{1}{Q_a}\right)^{\frac{1}{\varepsilon}} \left(\frac{Q_a}{c_{2,a}}\right)^{\frac{1}{\sigma}} \left[\mu_a \left(\frac{1}{C_a^w}\right)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\mu_a)\left(\frac{1}{C_a^m}\right)^{\frac{\varepsilon-1}{\varepsilon}}\right]$$
$$- \lambda_a p_2(1+\tau_2) = 0, \tag{35}$$

$$\frac{\partial V_a(\cdot)}{\partial h_{n,a}^w}: \quad (1-\omega)(1-\gamma_a)\alpha_n A_n \left(\frac{1}{Q_a}\right)^{\frac{1}{\varepsilon}} \left(\frac{Q_a}{c_{n,a}}\right)^{\frac{1}{\sigma}} \left(\frac{H_{n,a}}{h_{n,a}^w}\right)^{\frac{1}{\eta}} \left[\mu_a \left(\frac{1}{C_a^w}\right)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\mu_a)\left(\frac{1}{C_a^m}\right)^{\frac{\varepsilon-1}{\varepsilon}}\right]$$
$$- \lambda_a w^w z_a^w (1-\tau_h) - \beta \Lambda_{a+1}^w \underbrace{B_a z_a^w \nu (1-h_{n,a}^w - l_a^w)^{\nu-1}}_{\frac{\partial z_{a+1}^w}{\partial h_{n,a}^w}} = 0, \tag{36}$$

$$\frac{\partial V_a(\cdot)}{\partial h_{n,a}^m}: \quad (1-\omega)(1-\gamma_a)(1-\alpha_n)A_n \left(\frac{1}{Q_a}\right)^{\frac{1}{\varepsilon}} \left(\frac{Q_a}{c_{n,a}}\right)^{\frac{1}{\sigma}} \left(\frac{H_{n,a}}{h_{n,a}^m}\right)^{\frac{1}{\eta}} \left[\mu_a \left(\frac{1}{C_a^w}\right)^{\frac{\varepsilon-1}{\varepsilon}} + (1-\mu_a)\left(\frac{1}{C_a^m}\right)^{\frac{\varepsilon-1}{\varepsilon}}\right]$$
$$- \lambda_a w^m z_a^m (1-\tau_h) - \beta \Lambda_{a+1}^m \underbrace{B_a z_a^m \nu (1-h_{n,a}^m - l_a^m)^{\nu-1}}_{\frac{\partial z_{a+1}^m}{\partial h_{n,a}^m}} = 0, \tag{37}$$

$$\frac{\partial V_a(\cdot)}{\partial l_a^w}: \quad \alpha_l \left(\frac{1}{L_a}\right)\left(\frac{L_a}{l_a^w}\right)^{\frac{1}{\eta_l}} [\mu_a \theta^w + (1-\mu_a)\theta^m] - \lambda_a w^w z_a^w (1-\tau_h)$$
$$- \beta \Lambda_{a+1}^w \underbrace{B_a z_a^w \nu (1-h_{n,a}^w - l_a^w)^{\nu-1}}_{\frac{\partial z_{a+1}^w}{\partial l_a^w}} = 0, \tag{38}$$

$$\frac{\partial V_a(\cdot)}{\partial l_a^m}: \quad (1-\alpha_l)\left(\frac{1}{L_a}\right)\left(\frac{L_a}{l_a^m}\right)^{\frac{1}{\eta_l}} [\mu_a \theta^w + (1-\mu_a)\theta^m] - \lambda_a w^m z_a^m (1-\tau_h)$$
$$- \beta \Lambda_{a+1}^m \underbrace{B_a z_a^m \nu (1-h_{n,a}^m - l_a^m)^{\nu-1}}_{\frac{\partial z_{a+1}^m}{\partial l_a^m}} = 0, \tag{39}$$

where $\Lambda_a^g = \frac{\partial V_a(\cdot)}{\partial z_a^g} \equiv \lambda_a(1-\tau_h)w^g(1-h_{n,a}^g - l_a^g) + \beta\left[(1-\delta) + B_a(1-h_{n,a}^g - l_a^g)^\nu\right]\Lambda_{a+1}^g$, with $\Lambda_{31}^g = 0$.

Firm's Problem The minimization problem of the firms is

$$\min_{H_j^w, H_j^m} w^w H_j^w + w^m H_j^m, \tag{40}$$

subject to

$$H_j = \left[\alpha_j (H_j^w)^{\frac{\eta-1}{\eta}} + (1-\alpha_j)(H_j^m)^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}. \tag{41}$$

Thus:

$$\mathcal{L}(H_j^w, H_j^m) = w^w H_j^w + w^m H_j^m - \Omega_j \left[\left(\alpha_j (H_j^w)^{\frac{\eta-1}{\eta}} + (1-\alpha_j)(H_j^m)^{\frac{\eta-1}{\eta}}\right)^{\frac{\eta}{\eta-1}} - H_j\right]. \tag{42}$$

Taking first order conditions for $j = 1, 2$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial H_j^w} : w^w - \Omega_j \alpha_j \left(\frac{H_j}{H_j^w}\right)^{\frac{1}{\eta}} = 0, \tag{43}$$

$$\frac{\partial \mathcal{L}(\cdot)}{\partial H_j^m} : w^m - \Omega_j (1-\alpha_j) \left(\frac{H_j}{H_j^m}\right)^{\frac{1}{\eta}} = 0. \tag{44}$$

Rearranging,

$$H_j^w = H_j \left(\frac{\Omega_j \alpha_j}{w^w}\right)^{\eta}, \tag{45}$$

$$H_j^m = H_j \left(\frac{\Omega_j (1-\alpha_j)}{w^m}\right)^{\eta}. \tag{46}$$

Substituting these expressions for the gender total effective hours into the constraint of the firm (41)

$$H_j = \left[\alpha_j (H_j)^{\frac{\eta-1}{\eta}} \left(\frac{\Omega_j \alpha_j}{w^w}\right)^{\eta-1} + (1-\alpha_j)(H_j)^{\frac{\eta-1}{\eta}} \left(\frac{\Omega_j (1-\alpha_j)}{w^m}\right)^{\eta-1}\right]^{\frac{\eta}{\eta-1}} \Longrightarrow$$

$$\Omega_j = \left[\alpha_j^{\eta} (w^w)^{1-\eta} + (1-\alpha_j)^{\eta} (w^m)^{1-\eta}\right]^{\frac{1}{1-\eta}}. \tag{47}$$

This is the optimal cost per unit of total labor hours H_j. Therefore, the revenue from the firm is $p_j A_j H_j$, while the cost is $\Omega_j H_j$. Since from the optimality conditions marginal revenue equal marginal cost, then

$$p_j A_j = \Omega_j. \tag{48}$$

Thus, for sector 1, unit cost Ω_1 is equal to the price of sector 1 times the productivity of sector 1, i.e. $\Omega_1 = A_1$ since price of sector 1 is normalized to 1. Then, defining the gender wage ratio as $ratio_w \equiv (w^w/w^m)$ and dividing both FOCs of the firms

$$\begin{aligned} \frac{w^w}{w^m} &= \frac{\alpha_j}{(1-\alpha_j)} \left[\frac{H_j^m}{H_j^w}\right]^{1/\eta} \quad \Longrightarrow \\ \frac{H_j^m}{H_j^w} &= \left[ratio_w \frac{(1-\alpha_j)}{\alpha_j}\right]^{\eta}. \end{aligned} \tag{49}$$

Now, dividing the expression for total hours (41) by the women's total effective hours and using the expression before

$$\frac{H_j}{H_j^w} = (\alpha_j)^{\frac{\eta}{\eta-1}} \left[1 + \left(\frac{1-\alpha_j}{\alpha_j}\right)^{\eta} ratio_w^{\eta-1}\right]^{\frac{\eta}{\eta-1}}. \tag{50}$$

Substituting this into the first FOC of the firms

$$w^w = \Omega_j (\alpha_j)^{\frac{\eta}{\eta-1}} \left[1 + \left(\frac{1-\alpha_j}{\alpha_j}\right)^{\eta} ratio_w^{\eta-1}\right]^{\frac{1}{\eta-1}}. \tag{51}$$

From this expression I get the wage of the women. Using the gender ratio of wages, I also get the wage of the men ($w^m = \frac{w^w}{ratio_w}$). Finally, I use the expression for the unit cost (47) to get the price of sector 2.

A.3 Solution Algorithm

The stationary equilibrium is computed with a nested fixed-point algorithm. The outer loop searches for the gender wage ratio, the inner loop solves the household's life-cycle problem by backward induction, and a forward simulation aggregates the cohort and checks market clearing.

Outer loop: equilibrium wage ratio I guess the gender wage ratio $ratio_w \equiv w^w/w^m$. Since both market sectors operate under constant returns and p_1 is the numeraire, the firm's optimality conditions (43)–(44) reduce to pure price relations, so the remaining prices follow from $ratio_w$ alone. From $p_1 A_1 = \Omega_1$ I get $\Omega_1 = A_1$, which together with the women's wage expression yields w^w, then $w^m = w^w/ratio_w$, and finally $p_2 = \Omega_2/A_2$ using the unit cost (47). Taking these prices as given, I solve the household

problem, simulate a cohort, and aggregate effective hours and consumption. I then compute the firm's labor demand from the gender split (49) applied to $H_j = C_j/A_j$, and form the women's labor market gap

$$\text{gap}_w = \frac{H^w_{\text{demand}}}{H^w_{\text{supply}}} - 1, \qquad H^w_{\text{demand}} = \sum_j H^w_j, \qquad H^w_{\text{supply}} = \sum_{a=1}^{a=30} z^w_a h^w_a.$$

I update *ratio_w* by damped fixed-point iteration until $|\text{gap}_w| < \text{tol}$. The men's market then clears by Walras' law, so it isn't targeted separately.

Inner loop: single's problem (outside option) Before the couple's problem, I solve the single's life-cycle problem (18) for each gender. A single's only state is their own human capital, so I discretize z^g_a on a grid and solve by backward induction from $a = 30$ to $a = 1$, with $V^{g,S}_{31} = 0$: at each age and node I solve the single's first-order conditions, store the policy and the value $V^{g,S}_a(z^g)$, and interpolate across the grid. These values are fixed inputs to the couple's problem and do not depend on μ_a.

Inner loop: couple's problem and renegotiation I discretize the states (z^w_a, z^m_a, μ_a) on a grid and solve the married household's problem by backward induction from age 30 to age 1. At each age I solve the first-order conditions (33)–(39) together with the budget constraint; at the terminal age $\Lambda^g_{31} = 0$ and the problem is static, while at earlier ages the continuation value enters through Λ^g_{a+1} and the human-capital law of motion. This gives the allocation and each spouse's value of remaining married, $V^{g,M}_a$. I then compare it with the value of being single, $V^{g,S}_a$. If both spouses prefer to stay married, the weight is left unchanged. If one spouse would be better off single, the couple renegotiates: I move the bargaining weight in that spouse's favor until their married value just equals their single value, so they are again willing to remain. If no weight can satisfy both spouses, the marriage would dissolve, which does not occur on the calibrated equilibrium path. Since the weight may be renegotiated again later, each spouse's continuation value is taken at next period's renegotiated weight. I store the policies, the values, and the renegotiated weights across the grid.

Forward simulation and aggregation Starting from the initial states and bargaining weight $(z^w_{a=1}, z^m_{a=1}, \mu_{a=1})$, I simulate the representative cohort forward. At each age I carry over the weight from the previous age, solve the married problem, and apply the same check: if a spouse's married value falls short of their single value, I renegotiate the weight in their favor until the two are equal; otherwise the weight is unchanged. I use the resulting allocation, update human capital, and proceed to the next age. Aggregating effective hours $H^g = \sum_a z^g_a h^g_a$, output, and consumption over all ages gives the cohort totals, with which I verify market clearing and zero profits. The transfer T_a is set period by period from the government budget (17).